Week #13: [Orbital Anatomy]

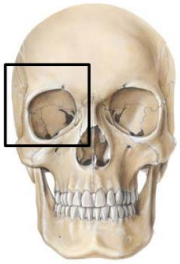
Orbital Basics
Orbital Muscles
Orbital Nerves
Arteries of Orbit

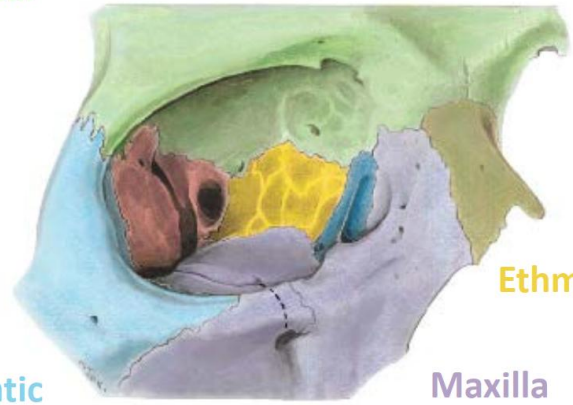
Orbital Basics

LO#1

Bony Orbit

- Many bones contribute
- Medial and inferior walls very thin



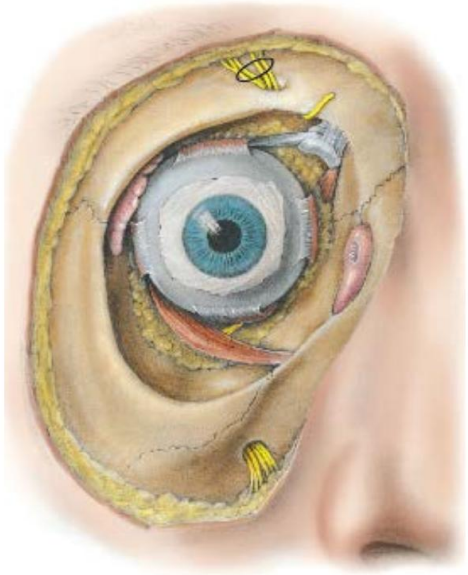


Also Palatine bone (not shown)

LO#3

Eye in Bony Orbit

- Eye is largely held in place by fat within orbit
- Muscles also help keep eye in place



BONY ORBIT pyramidal shaped bony structure housing the eye, comprised of seven bones: frontal, ethmoid, lacrimal, maxilla, zygomatic, sphenoid, palatine

a. ***superior wall (roof):*** frontal bone and lesser wing of sphenoid
b. ***medial wall*** – ethmoid bone (orbital plate, lamina papyracea), lacrimal bone, maxillary bone, sphenoid bone
c. ***inferior wall (floor)*** - maxillary, zygomatic, palatine (very tiny contribution to orbit) (*in a direct blow to the eye, the inferior wall is often damaged in a "blowout" fracture.)
d. ***lateral wall*** – zygomatic bone (frontal process), greater wing of sphenoid (* this is the strongest wall)

Fissures/foramina associated with the orbit & the structures transmitted through them.

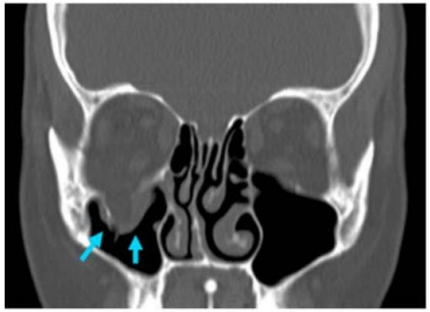
Fissure / foramina	Location	Transmits
Optic canal	at apex of orbit	optic nerve, ophthalmic a
Superior orbital fissure	between roof/lateral wall of orbit – this is the space formed between lesser & greater wing of the sphenoid bone	Superior & inf. ophthalmic veins, CN III, IV, VI, branches of V-1 (lacrimal, frontal, nasociliary nerves)
Inferior orbital fissure	between floor and lateral wall – formed by sphenoid bone	infraorbital artery, infraorbital branch of maxillary nerve (V-2)
Infraorbital foramen	In maxillary bone under orbit	infraorbital n & v
Supraorbital notch/foramen	In frontal bone above orbit	supraorbital nerve & vessels
Anterior/posterior ethmoidal foramen	In ethmoid bone	ethmoidal arteries and nerves

LO#1 & 3

Blowout Fractures

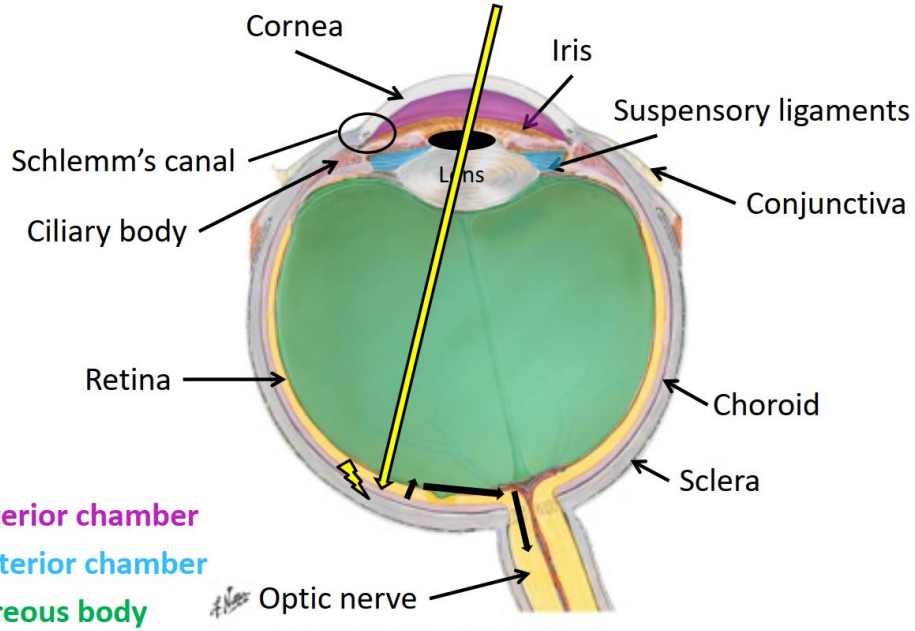
- indirect traumatic injury that displaces orbital walls
- medial and inferior walls of orbit are relatively weak
- inferior wall blowout fracture – fat from orbit often seen herniated into maxillary sinus "teardrop" sign





LO#4

Eyeball



EYEBALL - bulbus oculi - 3 major tunics - or layers of the eye

1. **Outer fibrous tunic**
    a. **sclera** - white of eye, posterior 5/6 of eyeball
    b. **cornea** - transparent anterior, allows light to enter eye, anterior 1/6 of eyeball
    c. **limbus** - junction between cornea & sclera
2. **Uvea (primarily a vascular tunic)**
    a. **choroid** - vascular layer, provides nutrients to outer retina
    b. **ciliary body** - complex of smooth muscles and vessels. Two functions: accommodation of lens and secretion of aqueous humor.

Ciliary muscle has the action of a sphincter. Receives *parasympathetic* innervation via CN III and ciliary ganglion. Upon contraction, it lessens the tension on fibers of the suspensory ligament of lens (lens becomes more rounded). Ability of lens to change shape is part of the accommodation (or near) reflex. Loss of this ability occurs with aging (presbyopia). By age 45 the ability of the lens to round up is lost.

The ***ciliary body*** secretes aqueous humor, which percolates into posterior chamber and subsequently to anterior chamber and is drained at the iridocorneal angle through this specialized trabecular meshwork into canal of Schlemm and ultimately into venous system. If rates of formation & absorption are out of balance, increased intraocular pressure results. This condition is ***glaucoma*** and if severe can lead to blindness.

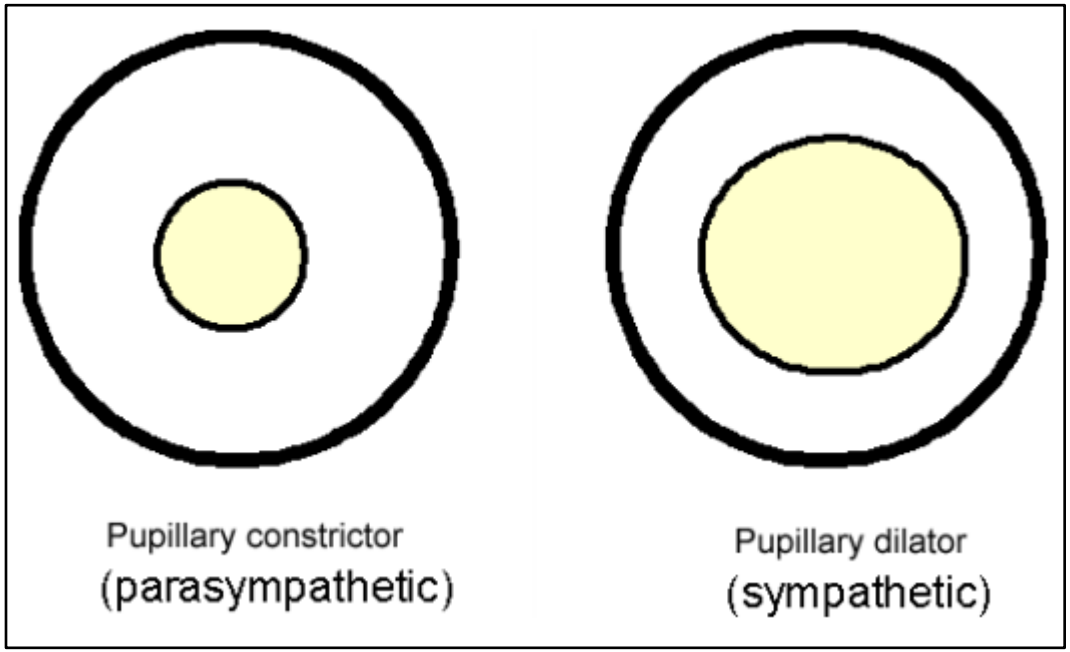
The ***iris*** - projects from the ciliary body, contains the central aperture, the ***pupil***, major bulk is a continuation of choroid. The pigmentation of iris determines eye color.

The iris contains 2 muscles:

1. The **sphincter pupillae** (constrictor) a circular muscle that alters the size of the pupil in response to light. It receives parasympathetics through the oculomotor

nerve by way of ciliary ganglion and short ciliary nerves. The iris controls the amount of light entering the eye.

2. The **dilator pupillae muscle** has radially arranged fibers innervated by the sympathetic system. (Post ganglionics from the superior cervical ganglion by the long ciliary nerves).

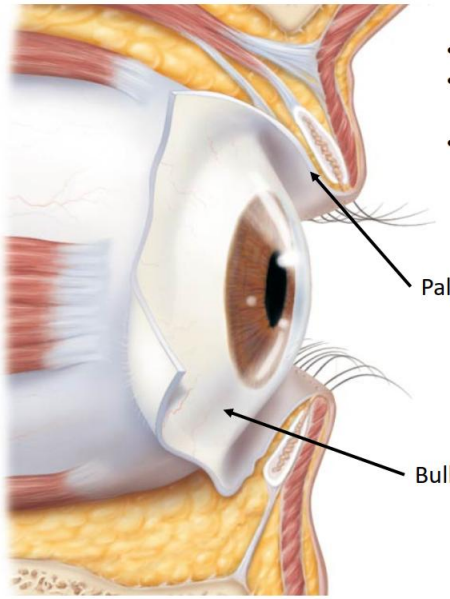


Tunica interna

a. **Retina - pars optica** is sensitive to light, the ten-layered retina is made of an outer pigmented epithelium and an inner nervous layer.
   i. **optic disk** - 1.5mm region where the optic nerve & central artery leave the eye (no photoreceptors here hence it is the "blind spot")
   ii. **macula lutea** - the 4.5mm central retinal area at posterior pole of eye, lateral to the optic disk.
   iii. Fovea centralis is the 1.5mm depressed area of the macula lutea. It is the area of **MOST ACUTE VISION.**

b. **Retina - pars cecum**
   a. Two parts:
      i. Iridial retina
      ii. Ciliary retina which are behind the iris and ciliary body respectively, **not light sensitive** (the “blind” part of the retina), ora serrata - point at which the ten-layered retina becomes a two layered ***ciliary epithelium***, continues on the posterior surface of the iris as the ***iridial epithelium***

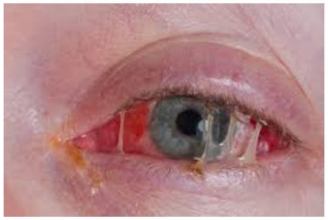
Conjunctival Sac



- Transparent mucous membrane
- Blood vessels can be seen beneath surface – "bloodshot eyes"
- Secretes lubricating mucous to keep eyes moist

LO#4

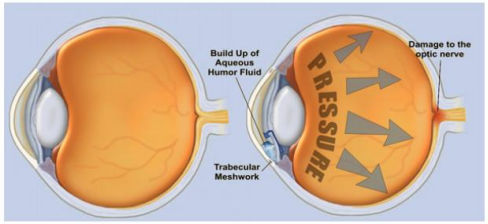
Pink Eye
- Infection of conjuctiva



Cataracts
- Loss of transparency of lens



Glaucoma
- Decreased drainage of aqueous humor
- Increases intraocular pressure
- Compression of neural layer of retina can lead to blindness



Muscles of the Iris

In dim light

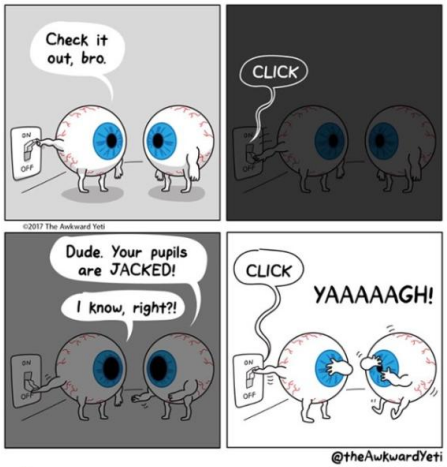
- SNS nerve fibres from superior cervical ganglion
- Innervate **Dilator pupillae**

Radially arranged dilator pupillae

Circularly arranged sphincter pupillae

In bright light

- PNS nerve fibres synapse carried with Oculomotor nerve
- Innervate **Sphincter pupillae**

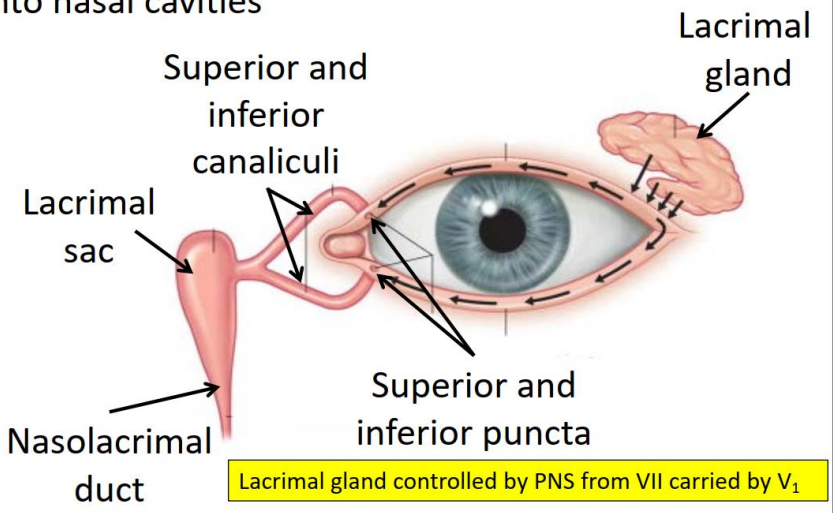


Try shining a light in a friend's eye to see the change in pupil diameter

LO#2

Lacrimal Apparatus

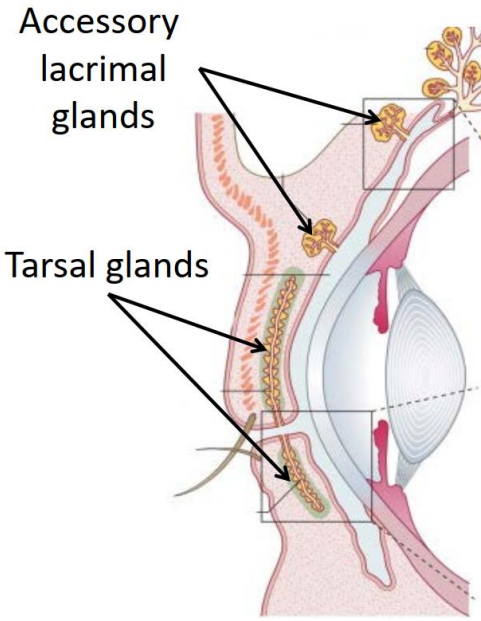
- Produces lacrimal fluid (tears)
- Blinking spreads fluid over cornea
- Drains into nasal cavities



Lacrimal gland controlled by PNS from VII carried by V_1

LO#2

Tarsal and Lacrimal Glands



- Lacrimal glands produce tears
- Tarsal glands in eyelids - produce oil
- Oil from tarsal gland helps keep tears adhered to cornea

LACRIMAL APPARATUS

- **Lacrimal gland** - lies in the upper, lateral part of orbit; supplied by lacrimal artery and lacrimal branch of V-1.
- **Lacrimal papilla** – swelling of the medial palpebral margins near the lacus; within the papilla sits a tiny opening: punctum lacrimale that is the opening of a tiny drainage canal: ***lacrimal canaliculus***.
  - Tears drain from the canaliculus into the lacrimal sac, then into the nasolacrimal duct, which empties into the inferior nasal meatus of the nasal cavity

LO#2

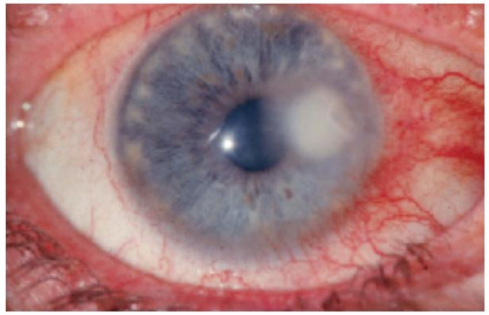
Sty

- Infected or clogged tarsal gland or duct



Corneal Ulcer

- Corneal irritation
- Can be caused by insufficient lacrimal fluid



Main Points 1 – Orbit basics

- Eye is protected by bony orbit
- Orbit is thin medially and inferiorly – thus vulnerable to damage
- Retina is a light sensitive layer of the eye that conveys signals to visual centres of brain via optic nerve
- Ciliary body controls the shape of the lens to allow accommodation
- Lacrimal fluid (tears) from lacrimal gland keep surface of eye moist

Orbital Muscles

LO#5

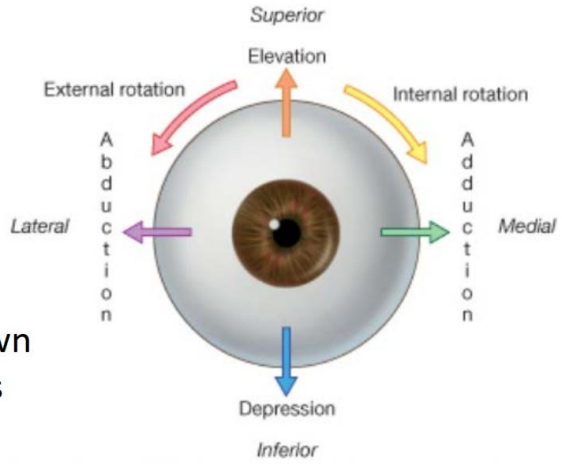
Muscles of the Orbit

7 extraocular muscles that move eye within orbit or elevate eyelid

- 1 eyelid muscle (levator palpebrae superioris)
- 4 recti (straight) muscles (superior, inferior, medial, lateral)
- 2 oblique muscles (superior and inferior)

LO#5

Movements of the Eye

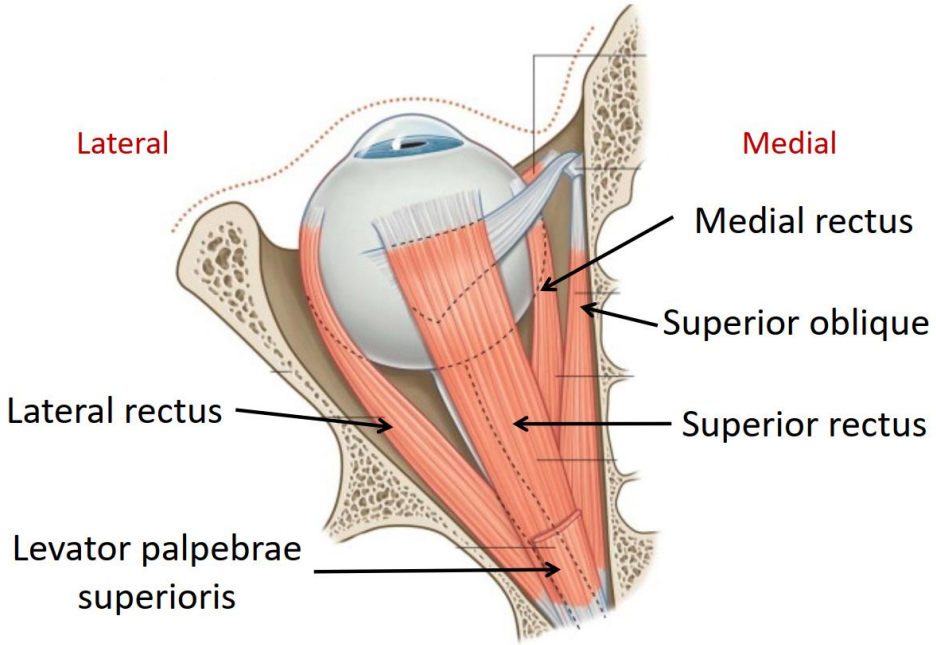


- Elevation – look up
- Depression – look down
- ADduction – eye looks towards nose
- ABduction – eye looks away from nose

EXTRAOCULAR MUSCLES (All are innervated by the oculomotor nerve (CNIII) except the superior oblique (CNIV) and lateral rectus (CNVI). **"(LR6SO4)3"**

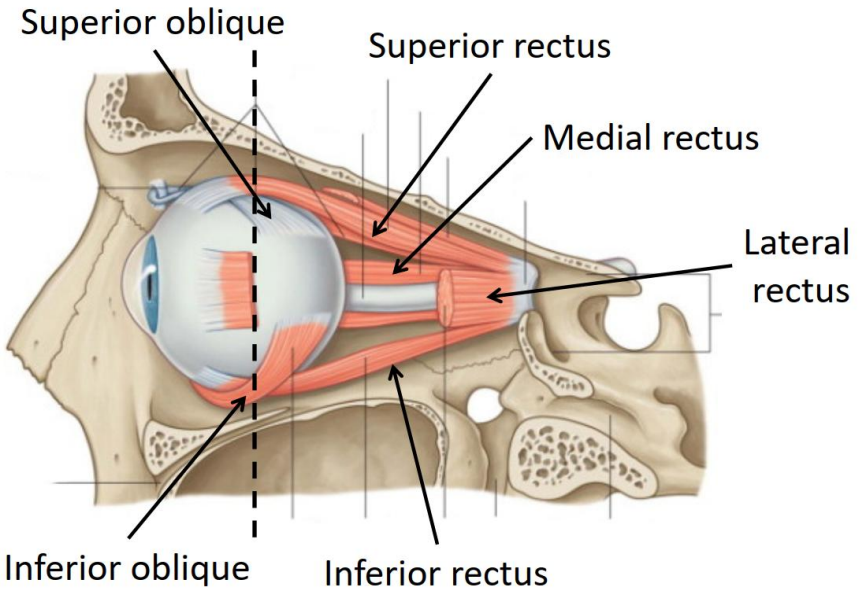
LO#5

Superior View of Extraocular Muscles



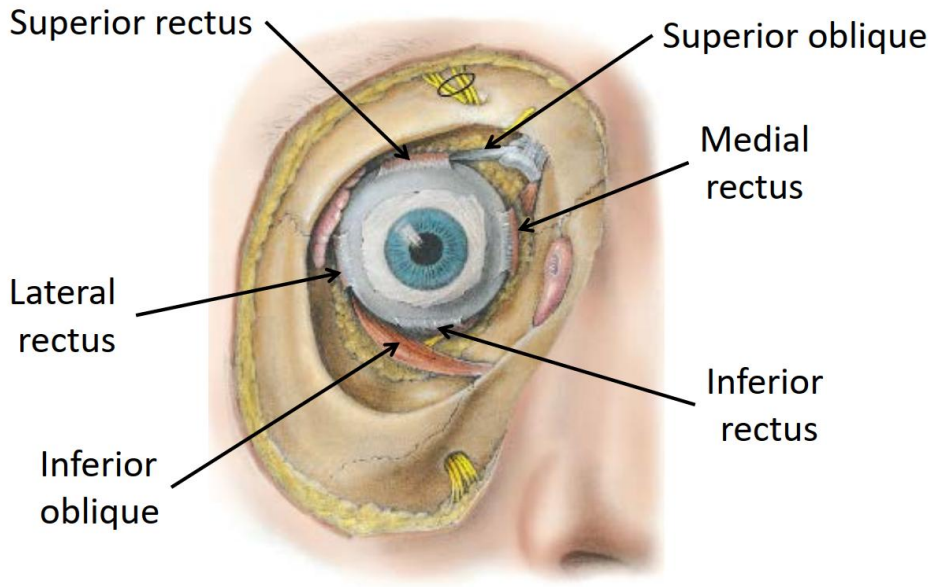
LO#5

Lateral View of Extraocular Muscles



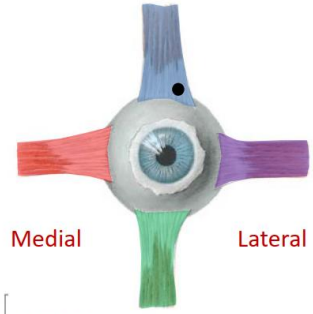
LO#5

Anterior View of Extraocular Muscles



LO#5

Recti Muscles

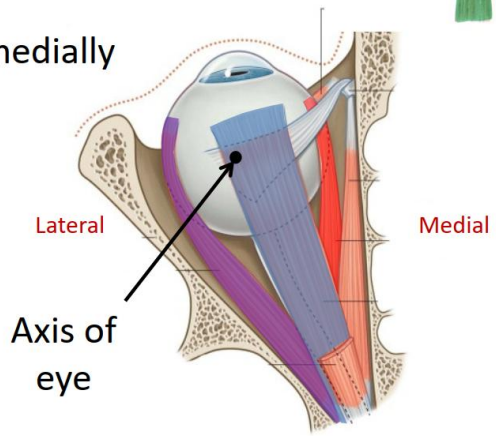


Superior Rectus
- eye looks up and medially

Inferior Rectus
- eye looks down and medially

Medial Rectus
- eye looks medially

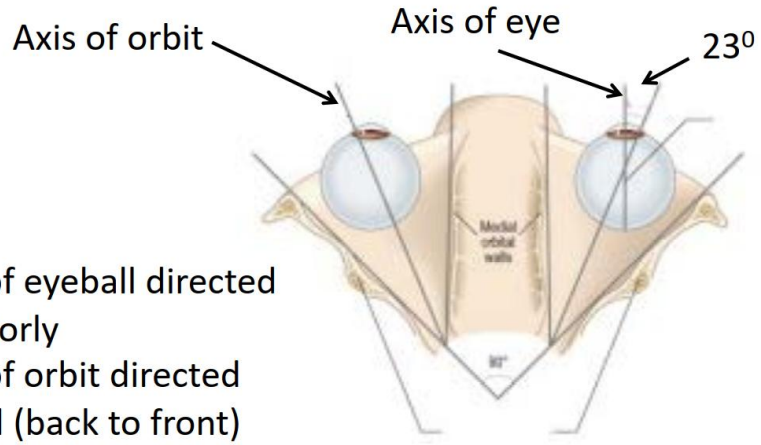
Lateral Rectus
- eye looks laterally



What are the actions of the recti muscles?

LO#5

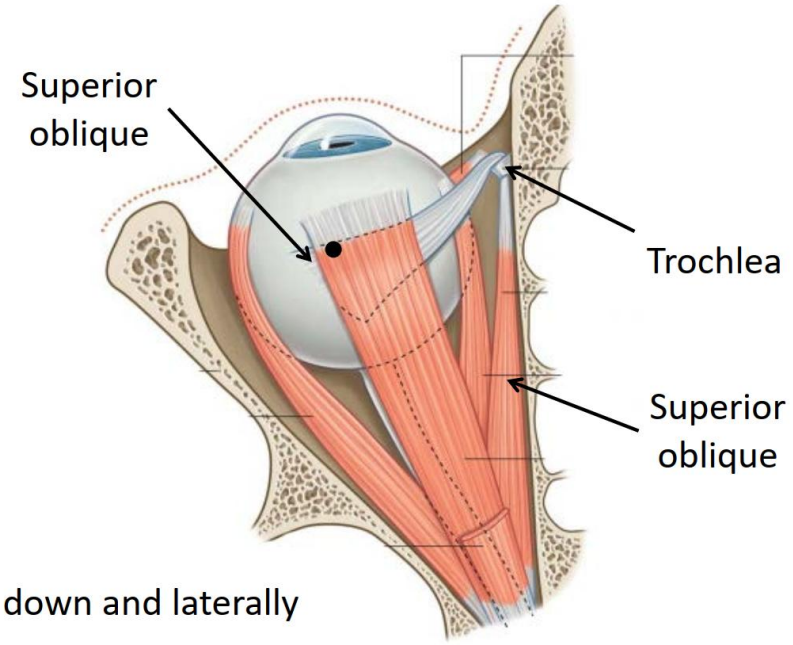
A Bit Tricky



- Axis of eyeball directed anteriorly
- Axis of orbit directed lateral (back to front)
- Affects line of pull of muscles

LO#5

Superior Oblique

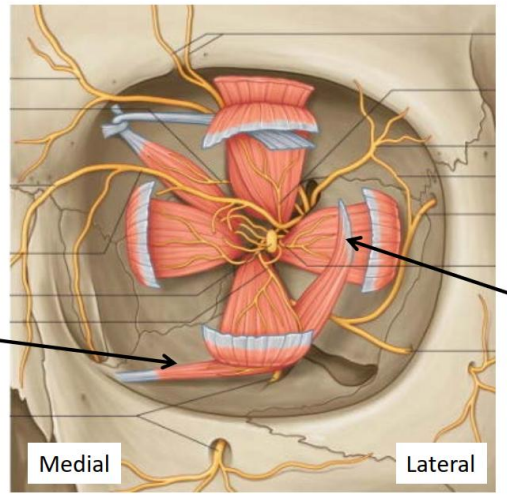


Action:
Eye looks down and laterally

LO#5

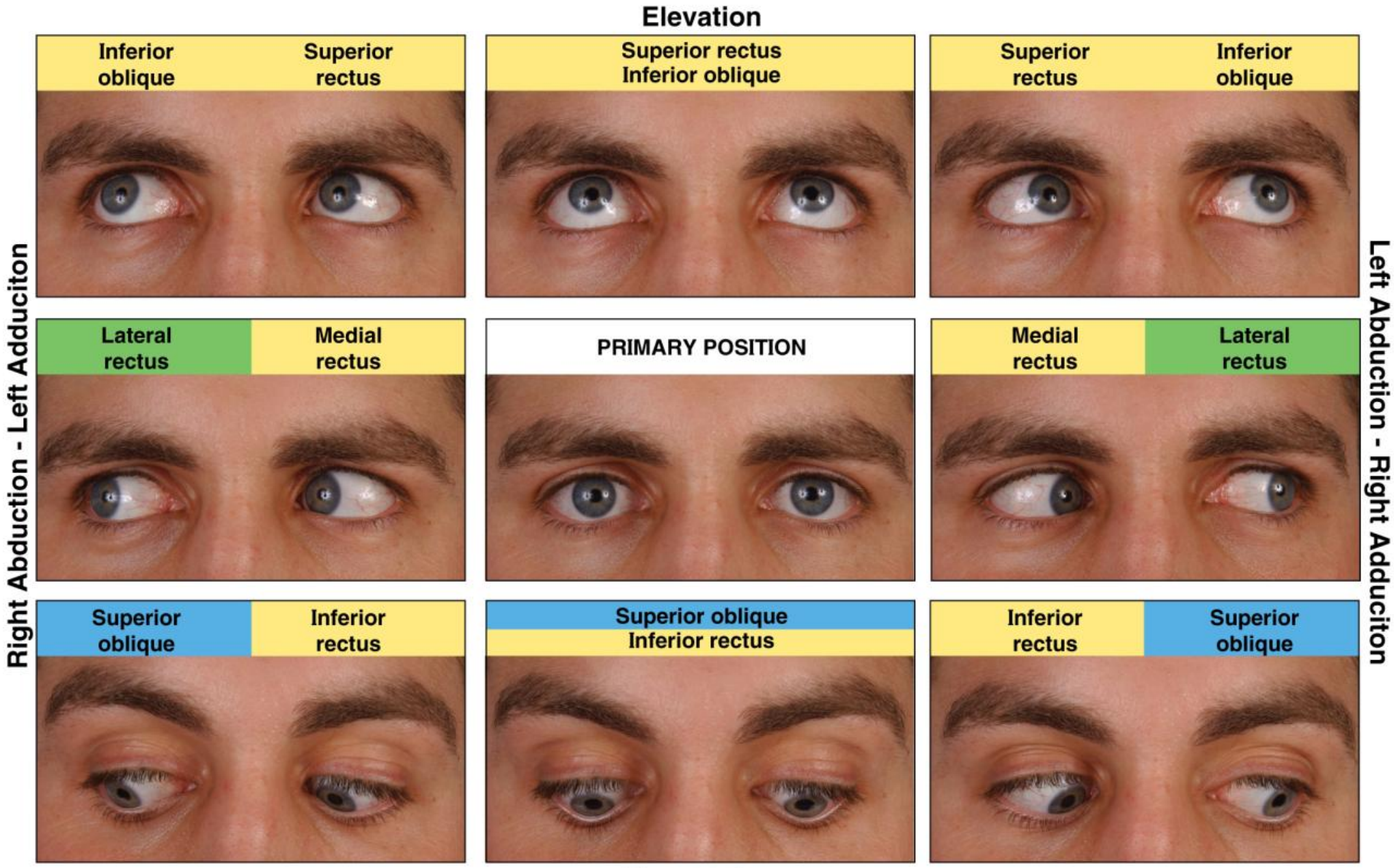
Inferior Oblique

Action:
Eye looks up and laterally



Correction note: Origin of inferior oblique is from **medial** aspect of orbit

Muscle	Origin	Insertion	Innervation	Function
Levator palpebra superioris (LPS)	Posterior orbit – above anulus tendineus	Superior tarsus (upper eyelid)	III	Elevate the eyelid
Lateral rectus	Anulus tendineus	Lateral side of eyeball, anterior to equator	VI	Direct the pupil laterally (abduct the pupil)
Medial rectus	Anulus tendineus	Medial side of eyeball, anterior to equator	III	Direct the pupil medially (adduct the pupil)
Superior rectus	Anulus tendineus	Superior side of eyeball, anterior to equator	III	Elevate the pupil, assist in turning pupil medially
Inferior rectus	Anulus tendineus	Inferior side of eyeball, anterior to equator	III	Depress the pupil, assist in turning pupil medially
Superior oblique	Posterior orbit, just below the LPS	Upper lateral quadrant of eyeball, posterior to the equator of the eye	IV	***DEPRESS*** the pupil; assist in turning pupil laterally
Inferior oblique	Floor of the orbit (from the nasal side, situated anteriorly)	Lower lateral quadrant of the eyeball, posterior to the equator of the eye	III	***ELEVATE*** the pupil; assist in turning the pupil laterally



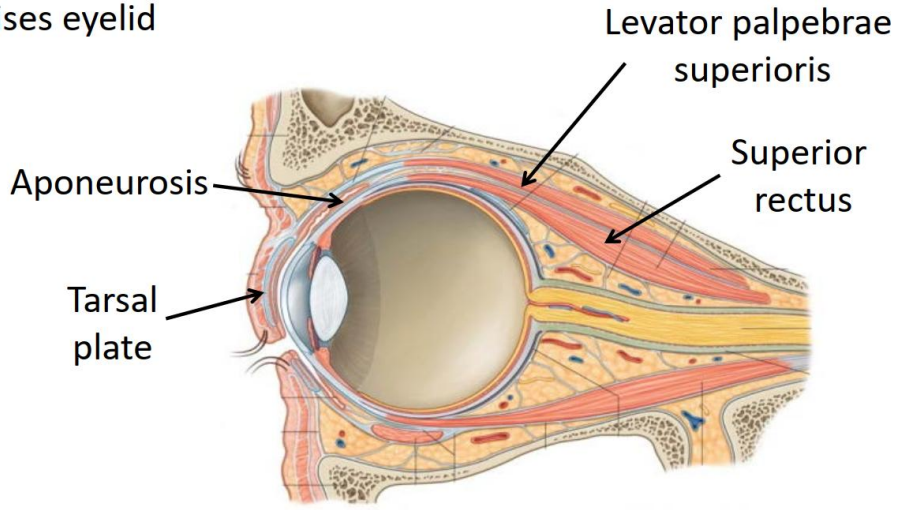
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LO#5

Levator palpebrae superioris

Action:
Raises eyelid



Main Points 2 – Extraocular Muscles

- 7 extraocular muscles move eye in orbit
- Movements produced by extraocular muscles
  - elevation, depression, aDduction, aBduction, internal and external rotation
- Axis of eye and axis of orbit are different
- Difference in orbit and eye axes affects line of pull of muscles

LO#6

Nerves of the Orbit

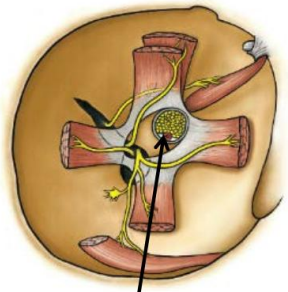
Sight
- Optic (CN II)

Motor to extraocular muscles
- Oculomotor (CN III)
- Trochlear (CN IV)
- Abducens (CN VI)

Sensory
- Ophthalmic branch of trigeminal (CN V_1)

LO#6

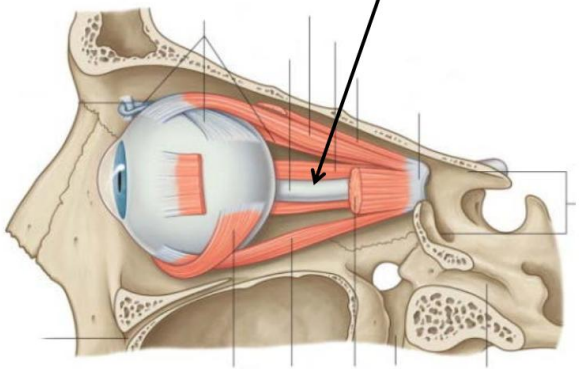
Sight – Optic Nerve (CN II)



Enters orbit through optic canal

Optic nerve

Optic nerve
(in dural sheath)



LO#6

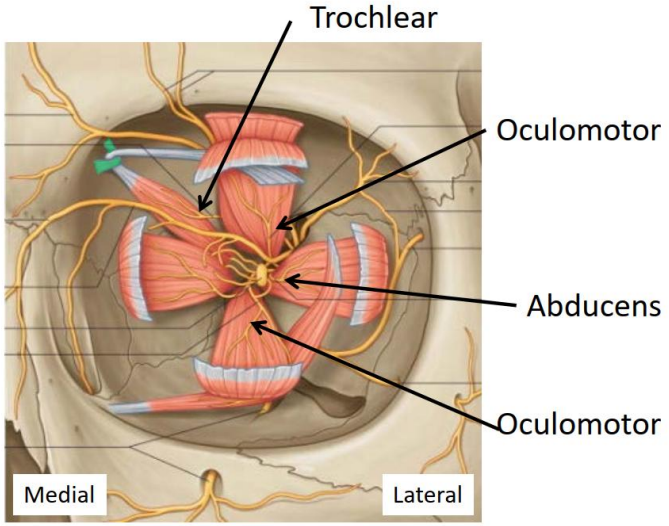
Motor Nerves of the Orbit (CN III, IV, VI)

Enter orbit through superior orbital fissure

Oculomotor
- levator palpebrae superioris
- superior rectus
- inferior rectus
- medial rectus
- inferior oblique

Trochlear
- superior oblique

Abducens
- lateral rectus



Note: Location of abducens nerve is correctly labelled in this diagram

V. NERVES – the eye is served by **6** (six) cranial nerves

1. **Optic nerve (CN II)** - mediates vision, sensory nerve, represents the axons of the ganglion cells of the retina. Nerve runs through the optic canal and joins the contralateral optic nerve forming the optic chiasm.
2. **Oculomotor nerve (CN III)** - motor innervation to: inferior oblique; superior, medial and inferior rectus muscles; levator palpebrae superioris; parasympathetic innervation (postganglionic) to ciliary muscle and sphincter pupillae muscle.
   - ***Parasympathetics:*** *preganglionic fibers synapse in the ciliary ganglion. The post- ganglionic fibers are carried to eyeball via the short ciliary nerves (which are actually branches of V1) to eventually innervate the ciliary muscle and the pupillary sphincter (constrictor).*
3. **Trochlear (cranial nerve IV)** motor to superior oblique
4. **Trigeminal (ophthalmic division)** cranial nerve V
   - ***frontal*** - supraorbital and supratrochlear branches
   - ***lacrimal nerve*** - to lacrimal gland
   - ***nasociliary*** - long & short ciliary, infratrochlear, ethmoidal (anterior & posterior)

**** Note** the structures running **through** the anulus tendineus include the oculomotor nerve (III), the nasociliary branch of V1, the abducens nerve (VI), the

optic nerve (II) and the ophthalmic artery. The trochlear nerve (IV), the frontal and lacrimal branches of V1 and the ophthalmic vein do not run through the anulus tendineus, but ***run above*** it.

LO#6

Abducens Nerve Palsy



- Unable to ABduct eye
- May be injured with increased intracranial pressure
- Characteristic head turn to avoid double vision

LO#6

Oculomotor Nerve Palsy



- Eye is in “down and out” position
- Ptosis (drooping of upper eyelid) of affected side

LO#6

Trochlear Nerve Palsy



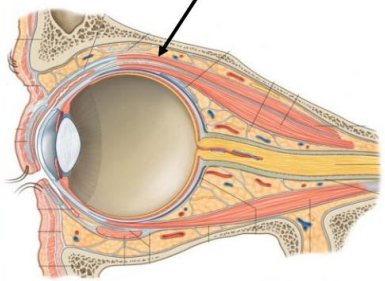


- Eye is in "up and in" position
- Commonly injured by trauma

Describe the presentation of this individual's eyes. Postulate the cause of the problem.

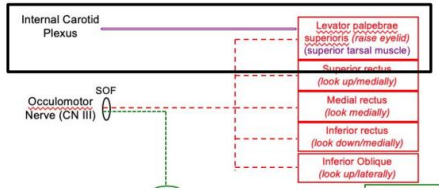


Levator palpebrae superioris
- Some fibres innervated by SNS
- (superior tarsal muscle)



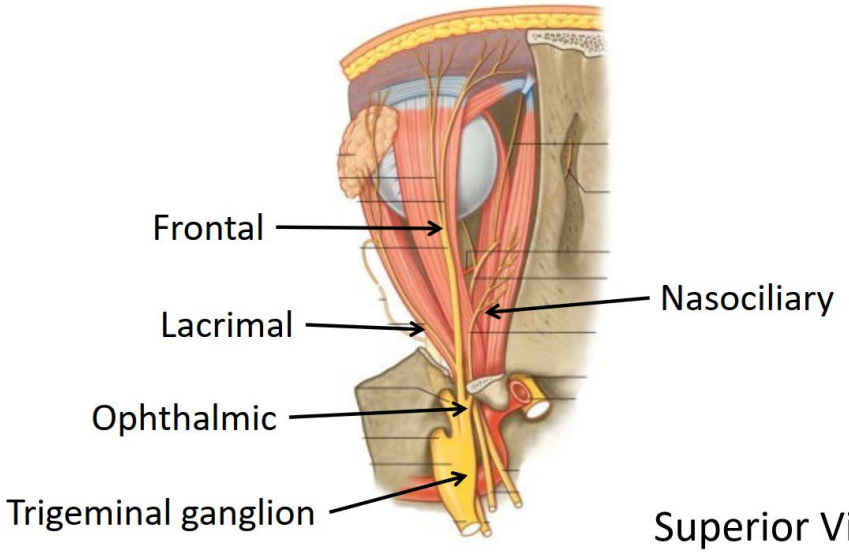
Horner's Syndrome: Interruption of superior cervical ganglion - leads to LACK of SNS on affected side

- Pupil: Miosis - constricted due to unopposed action of PNS
- Ptosis due to lack of SNS innervation to levator palpebrae superioris
- Red skin – vasodilation
- Anhydrosis – lack of sweating



LO#6

Sensory Nerve to Orbit
Ophthalmic Branch of Trigeminal Nerve (CN V_1)



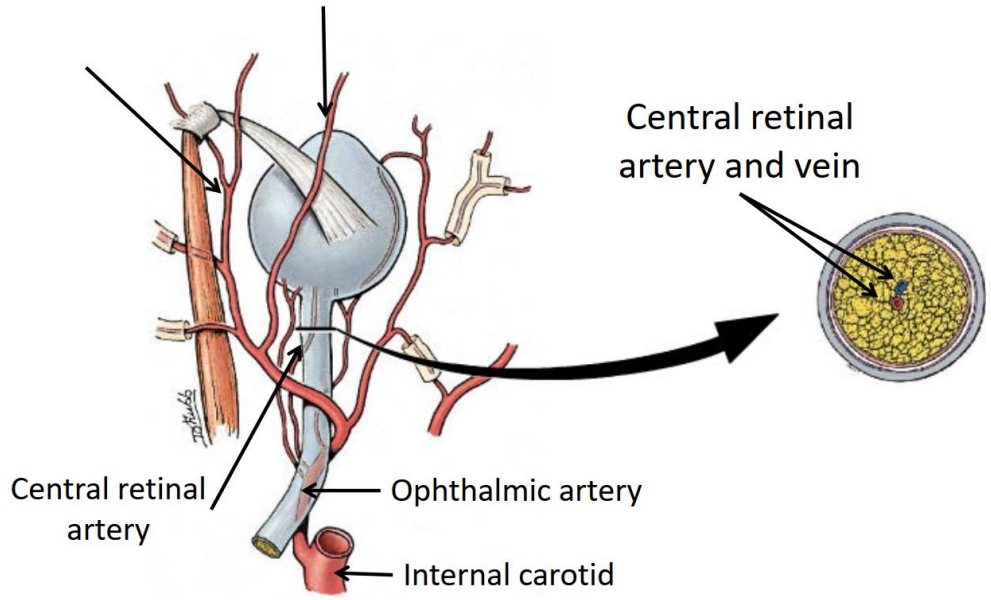
Superior View

Main Points 3 – Nerves of the Orbit

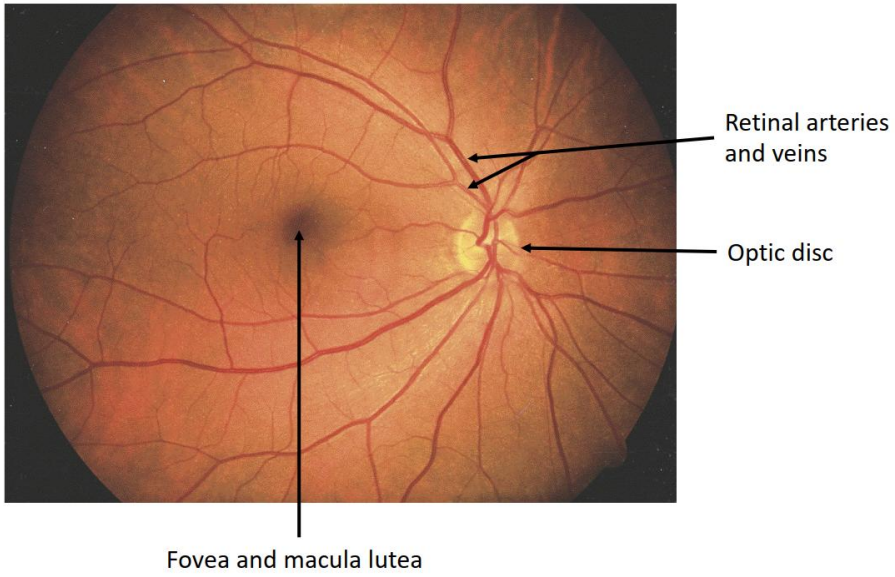
- Sight – optic nerve (CN II)
- Motor – oculomotor nerve (CN III)
  – trochlear nerve (CN IV)
  – abducens nerve (CN VI)
- General sensory – ophthalmic division of trigeminal nerve (CN V_1)

LO#7

Arteries of Orbit – Ophthalmic Artery



View of the Branches of the Ophthalmic Artery



LO#8

Relations of Cavernous Sinus

Optic chiasm (II)

Internal Carotid Artery

Oculomotor nerve (III)

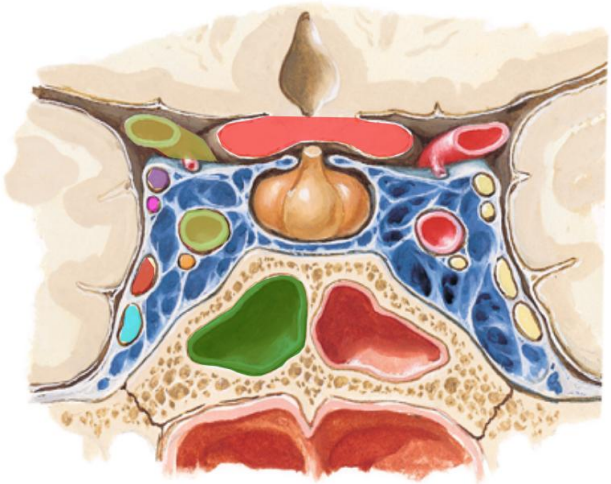
Trochlear nerve (IV)

Abducent Nerve (VI)

Ophthalmic nerve (V_1)

Maxillary nerve (V_2)

Sphenoidal sinus



VESSELS

Ophthalmic artery (br. of internal carotid), passes through the optic canal (foramen) within the dural sheath of the optic nerve. Gives off branches to orbital structures (see diagram, p. 8). One of its smallest branches but the most important is the Central Artery of the Retina. This artery pierces the optic nerve and runs within it to the optic disc. Spreads over the internal surface of the retina and supplies it. Obstruction of this artery leads to instant and complete blindness.

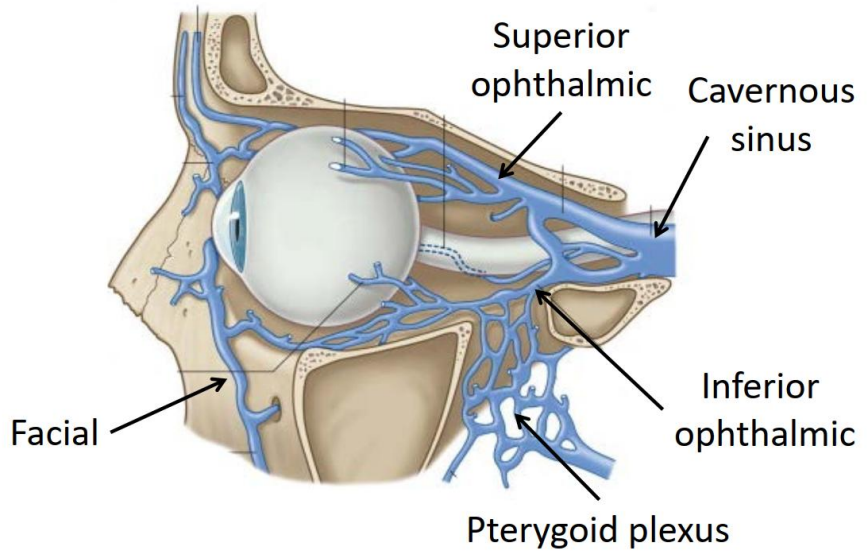
Ophthalmic veins there are two per eye (the superior and inferior) and they pass through the superior orbital fissure to vascularize the eyeball. The superior ophthalmic vein drains into the cavernous sinus; the inferior ophthalmic vein drains into the pterygoid venous plexus.

Refractory media - The transparent substances through which light travels to reach the retina.
This includes

- the ***cornea*** (anterior transparent continuation of sclera),
- the ***aqueous humor*** (fluid within the anterior and posterior chambers, synthesized by ciliary body),
- the ***lens*** (suspended from the ciliary body by zonular fibers, an opacity of the lens is a cataract),

LO#8

Veins of Orbit
Superior and Inferior Ophthalmic Veins



- the ***vitreous chamber*** (large area behind lens and posterior chamber filled with a gelatinous transparent mass called vitreous humor. The vitreous humor is not replaced. It has the consistency of raw egg whites, it is filled with collagen fibers and vitreocytes).

Main Points 4 – Vessels of the Orbit

- Ophthalmic artery supplies eye and structures in orbit
- Occlusion of central retinal artery results in blindness
- Ophthalmic veins have connection with cavernous sinus and facial veins
- Bacteria from face can enter cavernous sinus through this route
- Infection of cavernous sinus can affect motor and sensory nerves of orbit